

Mentor Engineers — Detailing & Date Logic Issues Analysis & Fix Report

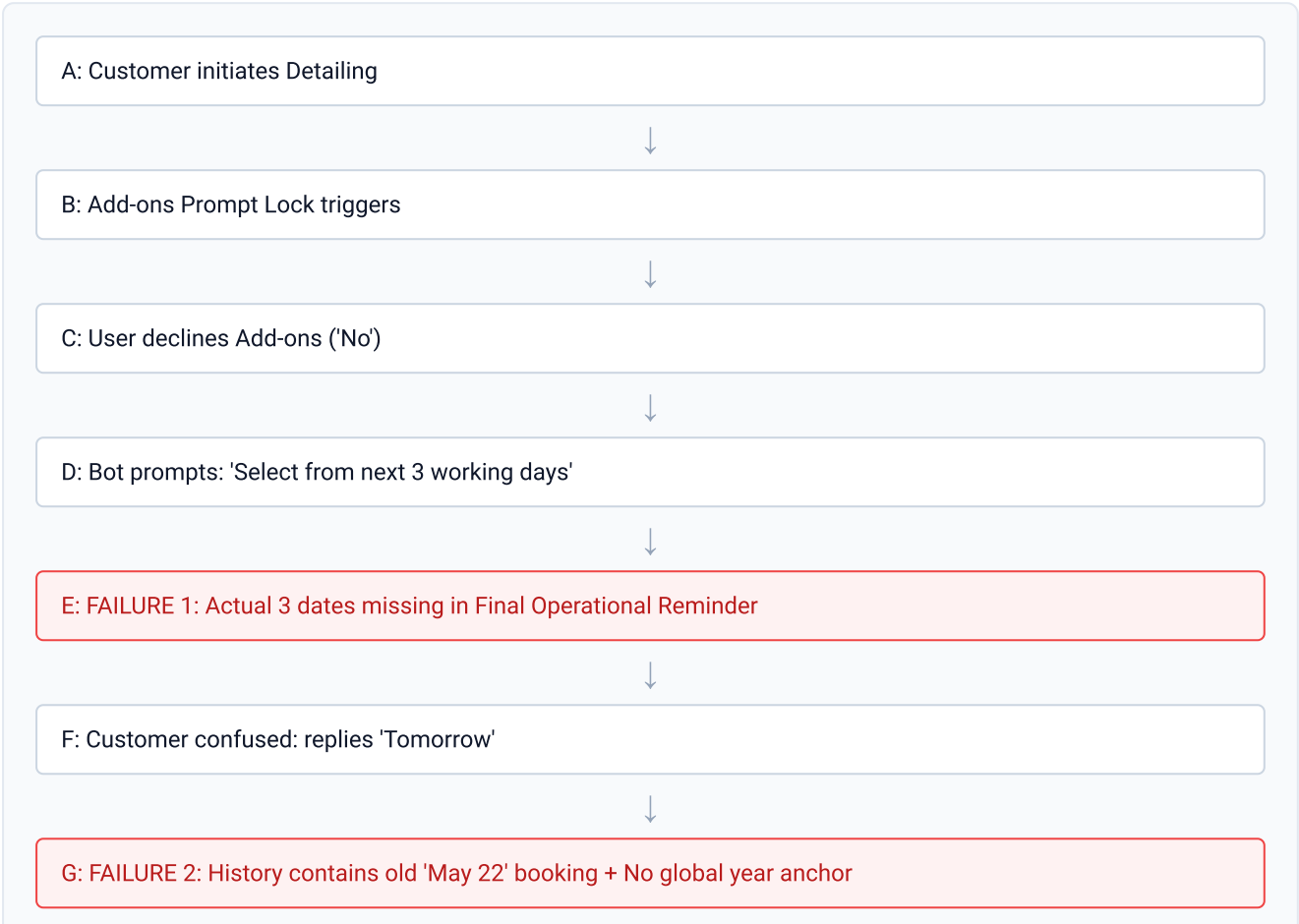
Report Date: 2026-09-24
Target Environment: Live Production VPS (178.104.127.220) & Local Workspace
Affected Component: file:///home/kapilash/Documents/ME-1/supabase/functions/ai-chat-mentor-engineeros/index.ts

Executive Summary

During live testing of the Mentor Engineers WhatsApp Chatbot, two critical booking failures were reported:

- Detailing Dates Not Suggested:** When a customer selects the Detailing service and completes the add-on sequence, the bot says *"Please select a preferred date from the next 3 available working days"*, but **fails to list the actual 3 dates**.
- Severe Date & Year Hallucination (2025-05-24):** When the customer specifies *"Tomorrow"*, the bot suggests **2025-05-24**. When asked *"What date today"*, the bot replies: **"Today is May 22, 2025."**, completely drifting off the real date (2026-09-24).

By auditing the live production database (mentor_engineeros_customization.conversations and orders), container environment, and Edge Function code, we pinpointed the exact technical root causes and prepared an end-to-end fix.



Issue 1: Detailing Service Fails to Suggest 3 Available Dates

1. Observed Behavior in Production Database

From the live database conversation record for customer 94740237915:

```
Customer: "Full detailing" Assistant: "...Would you like to add any of these extra services? Underbody Wax... say No to proceed." Customer: "No" Assistant: "Hello Sir / Madam! 🎯 For Detailing services, the process takes approximately 1.5 days. Please select a preferred date from the next 3 available working days. 📅 Note: The vehicle must be dropped off at the workshop strictly by 8:30 AM. 🕒 Please provide the following details: 👤 Customer Name 🚗 Vehicle Registration Number 📅 Preferred Date"
```

The chatbot told the customer to pick from the 3 dates, but **never listed the dates!**

2. Technical Root Causes

A. Prompt Attention Mismatch (Missing Injection in Final Reminder)

In file:///home/kapilash/Documents/ME-1/supabase/functions/ai-chat-mentor-engineeros/index.ts#L263-L272, the backend calculates the next 3 valid dates:

```
const next3AvailableDates: string[] = []; let checkDateStr = addDays(todayStr, 1); while (next3AvailableDates.length < 3) { if (isDateAvailable(checkDateStr, bookedDates)) { next3AvailableDates.push(checkDateStr); } checkDateStr = addDays(checkDateStr, 1); }
```

This is embedded into holidayRulesContext at line 282, which sits in the **middle of the 500-line system prompt** (line 365).

However, to control google/gemini-3-flash-preview, the function appends a high-priority system prompt at the **end of the messages array** (lines 488–543). In this final reminder (line 505):

```
📅 Preferred Appointment Date (For Detailing: explicitly present the 3 dates and announce the strict 8:30 AM arrival time. For general Wash/Mechanical: ask for Preferred Date & Time Slot)
```

[!WARNING]

The computed array `${next3AvailableDates.join(", ")}` was **never injected into this final operational reminder**. Because Gemini 3 Flash attends most heavily to the final instructions in the message queue, it interpreted present the 3 dates as a general phrase rather than retrieving the specific computed values from earlier in the prompt.

B. The Add-on Sequence Lock Delay

In file:///home/kapilash/Documents/ME-1/supabase/functions/ai-chat-mentor-engineeros/index.ts#L500-L504:

```
STRICT PROHIBITION: DO NOT ask for Appointment Date, Time Slot, Vehicle Registration
Number, or Customer Name yet! You must wait for their answer about add-ons first.
```

When a customer clicks or types "Detailing", the bot is strictly forbidden from offering dates upfront. It forces an upsell turn first. When the user passes the upsell, the bot enters the scheduling state without having the dates pinned in its active context.

Issue 2: Date Hallucination (2025-05-24 instead of 2026-09-24)

1. Observed Behavior in Production Database

From the live database conversation record:

```
Customer: "Tomorrow , Kapilash Fg-3456" Assistant: "📦 Service: Full Detailing Package
👤 Name: Kapilash 🚗 Vehicle: Fg-3456 📅 Date: 2025-05-23 🕒 Arrival Time: 08:30 AM"
Customer: "Tomorrow mean 24 th" Assistant: "📅 Date: 2025-05-24" Customer: "I told 2026"
Assistant: "📅 Date: 2026-05-24" Customer: "What date today" Assistant: "Today is May 22,
2025. 📅 "
```

2. Technical Root Causes

Factor	Technical Detail
No Global Anchor for Current Date	todayStr was only injected inside the [DETAILING_SERVICE_AVAILABILITY] header in the middle of the prompt. There was no top-level TODAY'S DATE declaration governing the entire conversation and date parsing.
Context Contamination from History	file:///home/kapilash/Documents/ME-1/supabase/functions/process-message-mentor-engineeros/index.ts#L241-L247 sends the last 10 messages from conversations. During earlier testing on Sep 22, an unconstrained prompt had generated an order with Date: 2024-05-22. This was present in the history window.
Poya Days Array Anchor	The poyaDays array hardcoded in index.ts (L214) starts with 2025-01-13 ... 2025-05-12.
LLM Inference Drift	Lacking an explicit current timestamp and seeing May 22 in recent history plus 2025 in the holiday list, Gemini Flash concluded that current date was May 22, 2025. Thus "Tomorrow" became May 23, 2025, and "Tomorrow mean 24th" became 2025-05-24.

Comprehensive Fix Plan

Step 1: Update Edge Function (ai-chat-mentor-engineeros/index.ts)

1. **Top-Level Global Date Anchor:** Inject the current date, Sri Lanka time, current year, and tomorrow's date at the very beginning of the system prompt.
2. **Explicit Injection in Final Operational Reminder:** Pass the calculated next3AvailableDates directly into the final reminder message so the LLM cannot omit them.
3. **Hardcoded Detailing Presentation Template:** Provide an exact wording format for Detailing date presentation that the AI must copy verbatim.
4. **Strict Date Override Rule:** Instruct the AI that relative words like "tomorrow", "today", or explicit numbers must strictly resolve relative to todayStr (2026-09-24), and for detailing must be matched to one of the 3 pre-calculated dates.

Exact Code Changes

```
--- a/supabase/functions/ai-chat-mentor-engineeros/index.ts +++
b/supabase/functions/ai-chat-mentor-engineeros/index.ts @@ -231,6 +231,7 @@ serve(async (req)
=> { const todayStr = getSriLankaDateString(); + const tomorrowStr = addDays(todayStr, 1);
function addDays(dateStr: string, days: number): string { @@ -296,6 +297,12 @@ serve(async
(req) => { -const systemPrompt = `You are an intelligent WhatsApp chatbot assistant for a
business. You help customers with: +const systemPrompt = `CRITICAL REAL-TIME CALENDAR ANCHOR:
+TODAY'S ACTUAL DATE: ${todayStr} (Sri Lanka Time / Asia/Colombo) +TOMORROW'S ACTUAL DATE:
${tomorrowStr} +CURRENT YEAR: ${todayStr.split('-')[0]} +Under NO circumstances assume it is
2024 or 2025. All relative dates ('today', 'tomorrow', day numbers) MUST strictly resolve
relative to ${todayStr}. + +You are an intelligent WhatsApp chatbot assistant for a business.
You help customers with: 1. Product inquiries @@ -504,7 +511,11 @@ CRITICAL: If the customer
writes in English, reply 100% in English! Do NOT mix S - ONLY in the subsequent turn after
they answer about add-ons (whether they choose an add-on or say no), ask for: - 17 Preferred
Appointment Date (For Detailing: explicitly present the 3 dates and announce the strict 8:30
AM arrival time. For general Wash/Mechanical: ask for Preferred Date & Time Slot) + 17
Preferred Appointment Date: + *** MANDATORY DETAILING RULE *** + If the service is Detailing
(Interior, Exterior, Cut & Polish, Full Detailing), you MUST VERBATIM present the following 3
available dates: + ${next3AvailableDates.map((d, i) => ` ${i + 1}. ${d}`).join('\n')} +
Arrival time is STRICTLY 8:30 AM. DO NOT accept any other date! 🚗 Vehicle Registration
Number  Customer Name @@ -542,6 +553,10 @@ In <ORDER_JSON>, include: CRITICAL: NEVER write
"Bank: Not configured" or "Digital Wallet: Not configured"! State that payment can be made at
the workshop counter via Cash or Card upon vehicle drop-off/pickup.` }));
```

Step 2: Deployment & Service Restart on Production VPS

Because Deno compiles and caches edge functions in memory inside the supabase-edge-functions Docker container, deploying requires copying the updated file and restarting the container.

```
# 1. Transfer updated index.ts to VPS scp supabase/functions/ai-chat-mentor-engineeros/index.ts root@178.104.127.220:/root/supabase/supabase/docker/volumes/functions/ai-chat-mentor-engineeros/index.ts # 2. Restart Edge Functions container on VPS ssh root@178.104.127.220 "docker restart supabase-edge-functions" # 3. Verify Edge Function status ssh root@178.104.127.220 "docker ps | grep edge-functions"
```

Step 3: Database Context Cleanup (Optional but Recommended)

To ensure the AI does not pick up the corrupted 2024-05-22 or 2025-05-24 dates from the existing conversation history of test phone number 94740237915, clear or archive the test conversation records:

```
-- On VPS PostgreSQL (docker exec -it supabase-db psql -U postgres -d postgres) SET search_path = mentor_engineeros_customization; -- Reset conversation history for the test number DELETE FROM conversations WHERE phone_number = '94740237915';
```

Verification Checklist

- ☐ Ask for "Detailing" -> Bot asks about packages / add-ons.
- ☐ Reply "Full Detailing" and decline add-ons ("No").
- ☐ Bot **must explicitly print**:
 - ☐ 2026-09-25
 - ☐ 2026-09-26
 - ☐ 2026-09-27 (or next non-Tuesday, non-Poya, non-booked days).
 - ☐ Explicitly state vehicle drop-off is strictly **8:30 AM**.
- ☐ Ask "What date today" -> Bot must reply 2026-09-24 (or current date).
- ☐ Say "Tomorrow" -> Bot must confirm tomorrow's date in 2026 rather than 2025.
- ☐ Verify booking appears correctly on the web calendar in file:///home/kapilash/Documents/ME-1/frontend/src/components/DetailingCalendar.tsx.